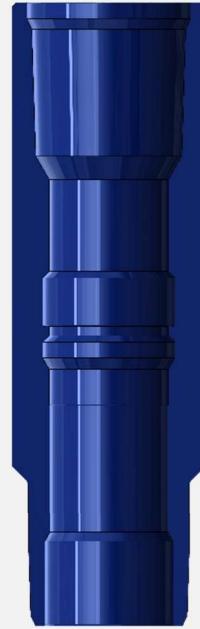
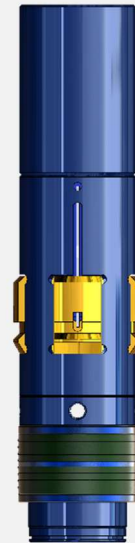


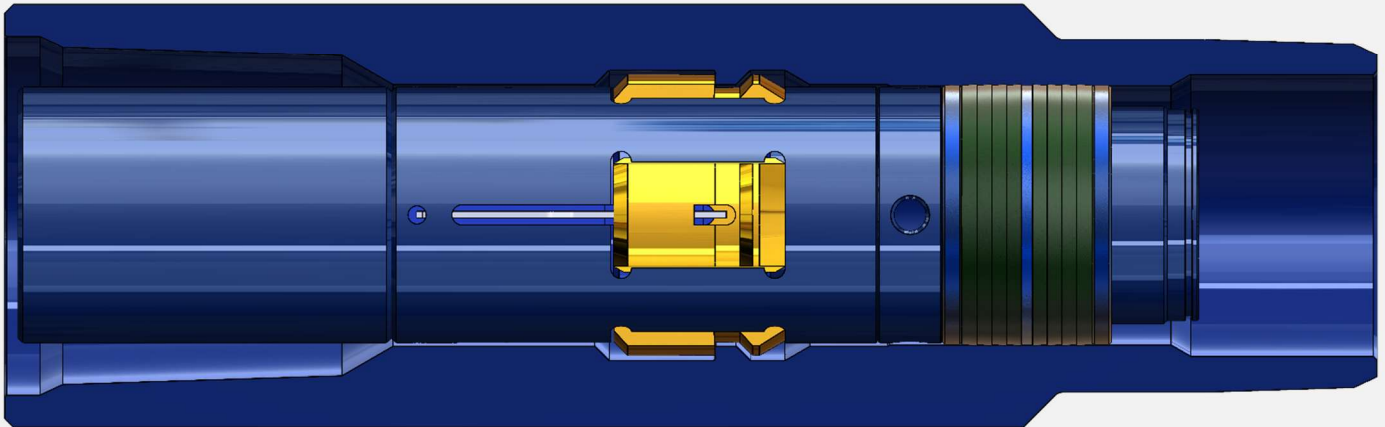
SCUBE INDUSTRIES – X Landing Nipples & Lock**Mandrels**

- Otis X landing nipples are selective-type nipples.
- Provide predefined landing and locking locations for subsurface flow control equipment.
- Landing nipples are manufactured with a common internal profile to ensure compatibility with standard Otis X lock mandrels and flow control devices.
- Designed for use in standard-weight tubing.
- Landing nipples with identical internal diameters may be included in the completion string in any sequence.
- Provides flexibility for locating, setting, and repositioning subsurface flow control equipment along the tubing string.

SCUBE INDUSTRIES – XN Landing Nipples & Lock Mandrels

- Otis XN no-go landing nipples are intended for single-nipple applications.
- Can also be used as the bottom nipple in a series of Otis X or R landing nipples.
- For a given tubing size and weight, all Otis XN landing nipples share a common packing bore internal diameter.
- Common packing bore ensures consistent tool compatibility.
- Otis XN landing nipples are designed for installation in standard-weight tubing.

**LANDING NIPPLE****LOCK MANDREL**



FEATURES

Landing Nipples

- Large-bore internal design minimizes flow restriction and pressure drop
- Universal landing nipple with a single internal profile, compatible with standard locking mandrels

Lock Mandrels

- Retractable locking keys for reliable setting and retrieval
- Locking mechanism designed to withstand differential pressure from both above and below
- Extra-large internal diameter (ID) to support higher flow rates and improved production efficiency

BENEFITS

Landing Nipples

- High versatility reduces completion and production maintenance costs
- Simple, reliable operation
- Flexible options for running, setting, and retrieving subsurface flow-control equipment

Lock Mandrels

- Faster slickline operations enabled by retractable locking keys
- Positive operator control for accurate locating, landing, and locking
- Internal fishing neck with large ID maximizes flow capacity

APPLICATIONS

- Supports gauge hanger installation for bottomhole pressure and temperature (P/T) surveys.
- Provides a positive locator interface for straddle system deployment and positioning.
- Enables plugging operations under pressure to minimize well intervention downtime.
- Allows near-unlimited depth selection for setting and locking subsurface flow control equipment.

Otis X and XN landing nipples and lock mandrels

TUBING								FOR STANDARD TUBING WEIGHTS						LOCK MANDREL ID	
								‘X’ PROFILE		‘XN’ PROFILE					
SIZE		WEIGHT		ID		DRIFT				PACKING BORE		PACKING BORE			
IN.	MM	LB/FT	KG/M	IN.	MM	IN.	MM	IN.	MM	IN.	MM	IN.	MM	IN.	MM
1.660	42.16	2.3	3.42	1.380	35.05	1.286	32.66	1.250	31.75	1.250	31.75	1.135	28.83	0.62	15.75
		2.4	3.57												
1.900	48.26	2.4	3.57	1.660	42.16	1.516	38.51	1.500	38.10	1.500	38.10	1.448	36.78	0.75	19.05
		2.76	4.11	1.610	40.89										
		2.9	4.32												
2.063	52.40	3.25	4.84	1.751	44.48	1.657	42.09	1.625	41.28	1.625	41.28	1.536	39.01	0.75	19.05
2 3/8	60.33	4.6	6.85	1.995	50.67	1.901	48.29	1.875	47.63	1.875	47.63	1.791	45.49	1.00	25.40
		4.7	6.99												
2 7/8	73.03	6.4	9.52	2.441	62.00	2.347	59.61	2.313	58.75	2.313	58.75	2.205	56.01	1.38	35.05
		6.5	9.67												
3 1/2	88.90	9.3	13.84	2.992	76.00	2.867	72.82	2.813	71.45	2.813	71.45	2.666	67.72	1.75	44.45
		10.2	15.18	2.992	74.22	2.797	71.04	2.750	69.85	2.750	69.85	2.635	66.93		
4	101.60	11	16.37	3.476	88.29	3.351	85.10	3.313	84.15	3.313	84.15	3.135	79.63	2.12	53.85
4 1/2	114.30	12.75	18.97	3.958	100.53	3.833	97.36	3.813	96.85	3.813	96.85	3.725	94.62	2.62	66.55
5	127.00	13	19.35	4.494	114.14	4.369	110.97	4.313	109.55	4.313	109.55	3.987	101.27	2.62	66.55
5 1/2	139.70	17	25.30	4.892	124.26	4.767	121.08	4.562	115.87	4.562	115.87	4.455	113.16	3.12	79.25